

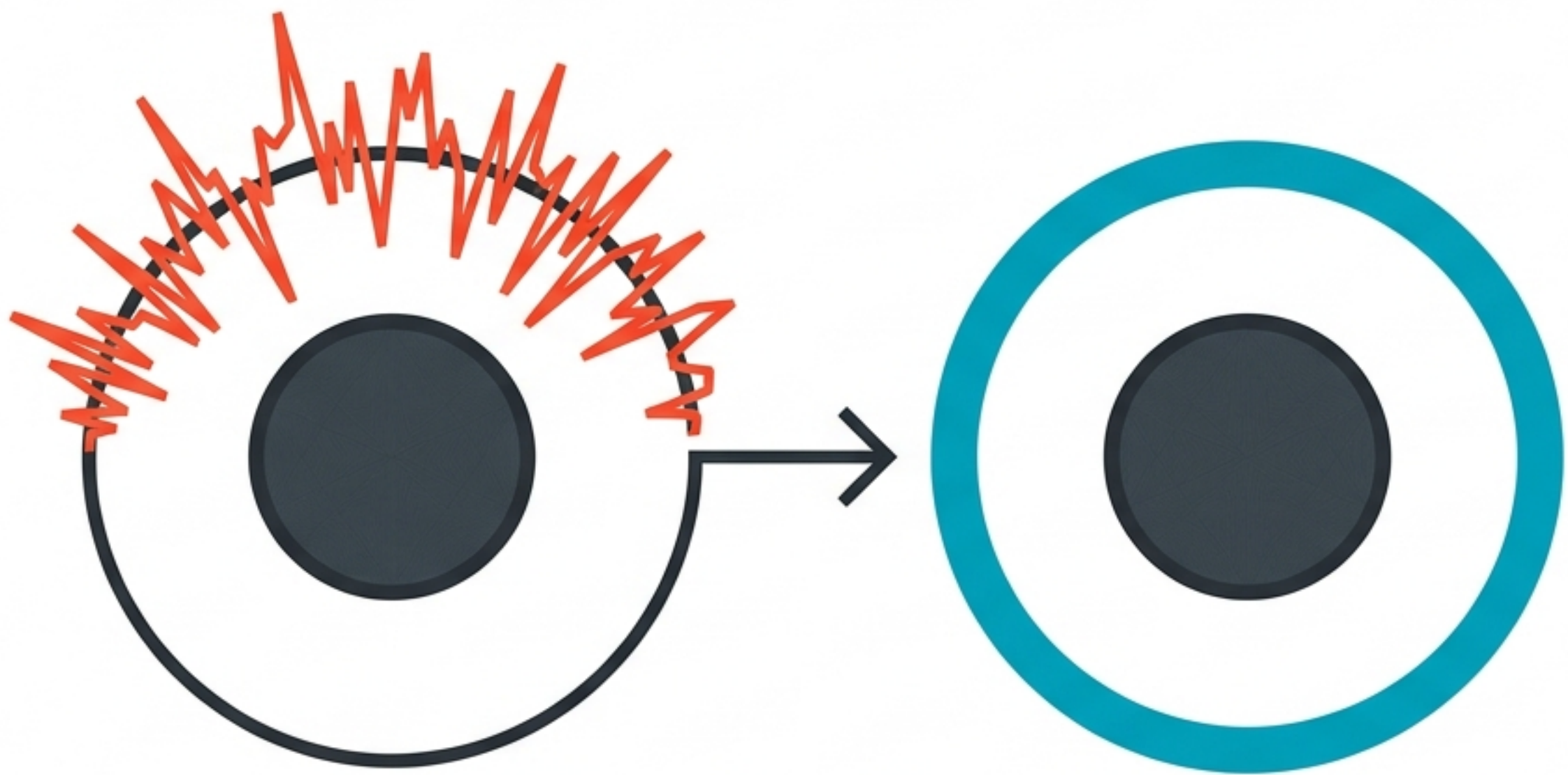
Thompson's Group F is Nonamenable

From a 100-year-old paradox to a machine-verified mathematical breakthrough.

Explainer Deck / Source: OpenAI

(Sept 23, 2026)

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Groups as symmetries, and the search for an average.

In mathematics, an invariant mean is a way to calculate an average over a topological group that remains completely unchanged even if you shift your perspective.

The Rules of the Mean M

Normalized: $M(1) = 1$

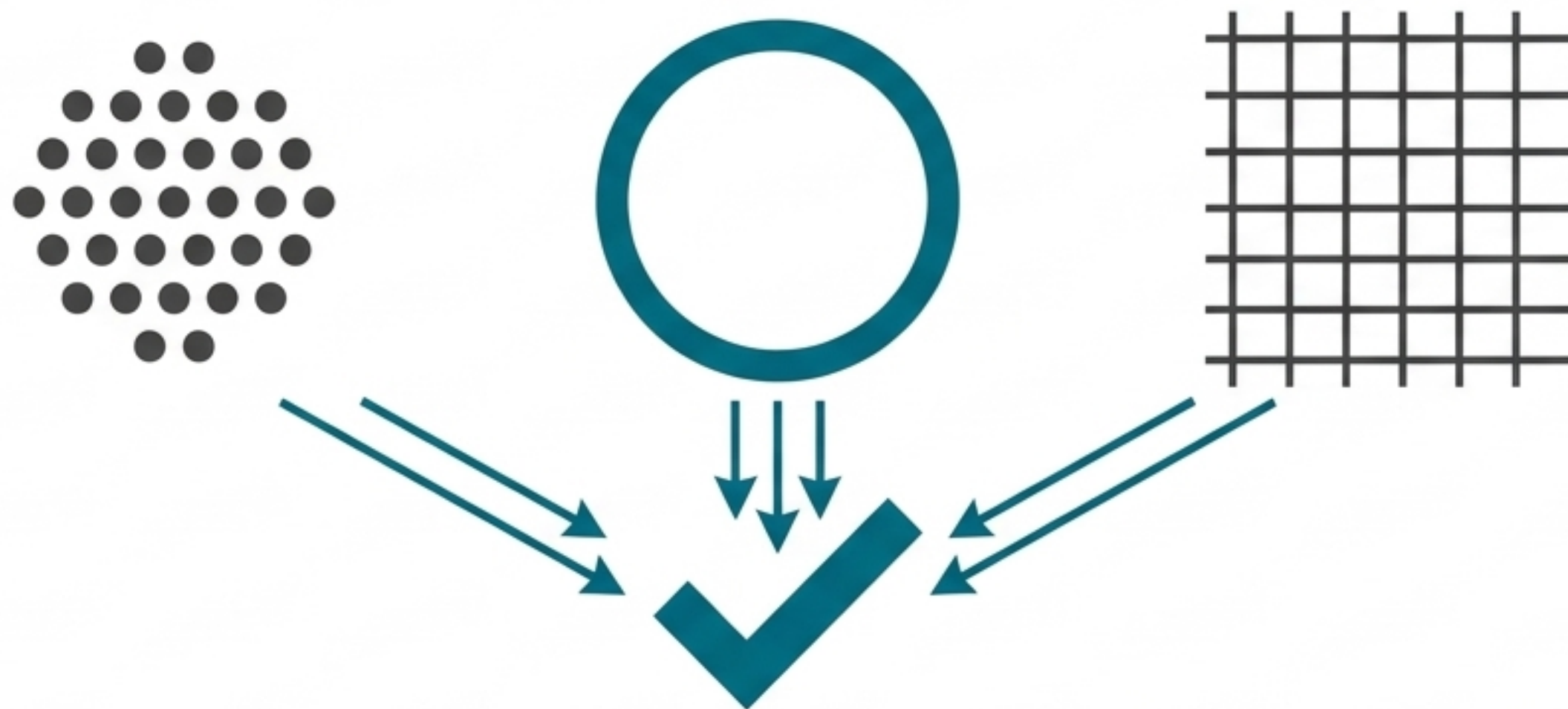
Positive: $M(\phi) \geq 0$ for $\phi \geq 0$

Invariant: $M(\text{shifted } \phi) = M(\phi)$

Amenability is the existence of harmony.

A discrete group is amenable if its bounded real functions admit a left-invariant mean.
Introduced by John von Neumann in 1929 in response to the Banach-Tarski paradox, amenability implies a group does not allow paradoxical, volume-multiplying geometric decompositions.

$$M(g \mapsto \varphi(hg)) = M(\varphi)$$

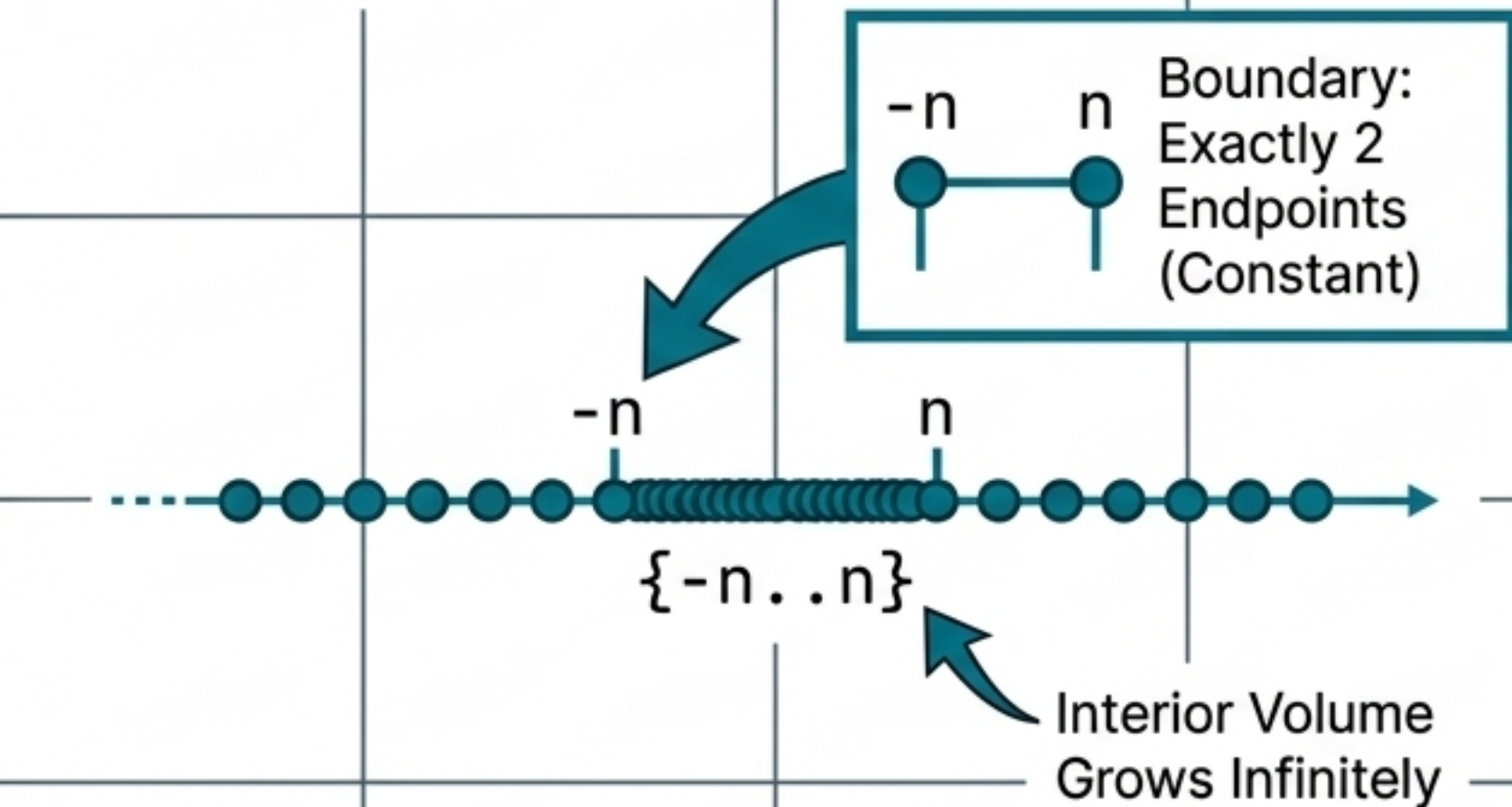


Finite groups, abelian groups, and solvable groups are universally amenable.

The Følner condition translates algebra into geometry.

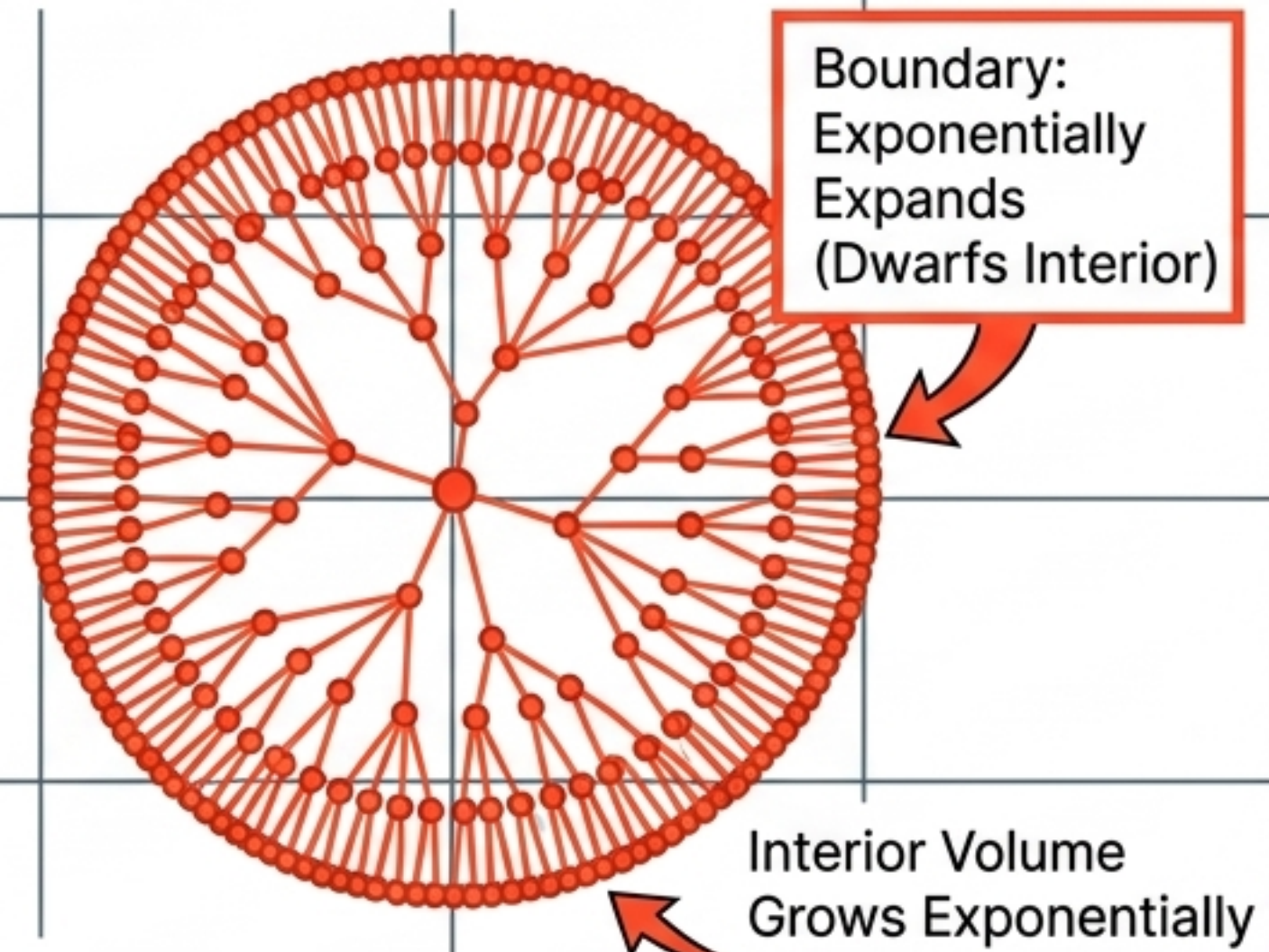
If a group is amenable, it must have finite sets where the boundary (elements shifted out of the set) is vanishingly small compared to the total volume.

Amenable ($\mathbb{Z}$)



$$\text{Ratio: } \frac{2}{2n+1} \rightarrow 0$$

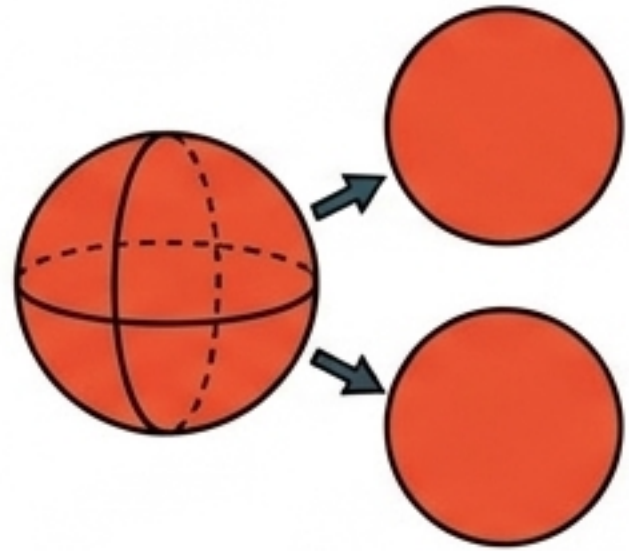
Nonamenable (F_2)



$$\text{Ratio: } \frac{2 \cdot 3^n}{2 \cdot 3^n - 1} \approx 1$$

A century of searching for mathematical monsters.

For decades, mathematicians wondered if the free group F_2 was the only reason a group could be nonamenable. Once proved false, the hunt for edge-cases accelerated.

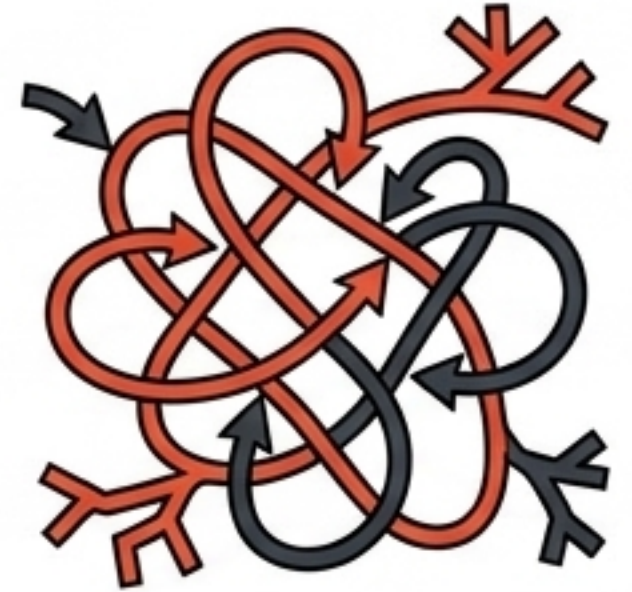


1929

John von Neumann defines **measurable (amenable)** to isolate groups that prevent this geometric paradox.

1949

The von Neumann-Day Question: Must a **nonamenable** group contain a free subgroup F_2 ?



1924

Banach-Tarski Paradox:
A sphere cut and reassembled into two identical spheres.

1949

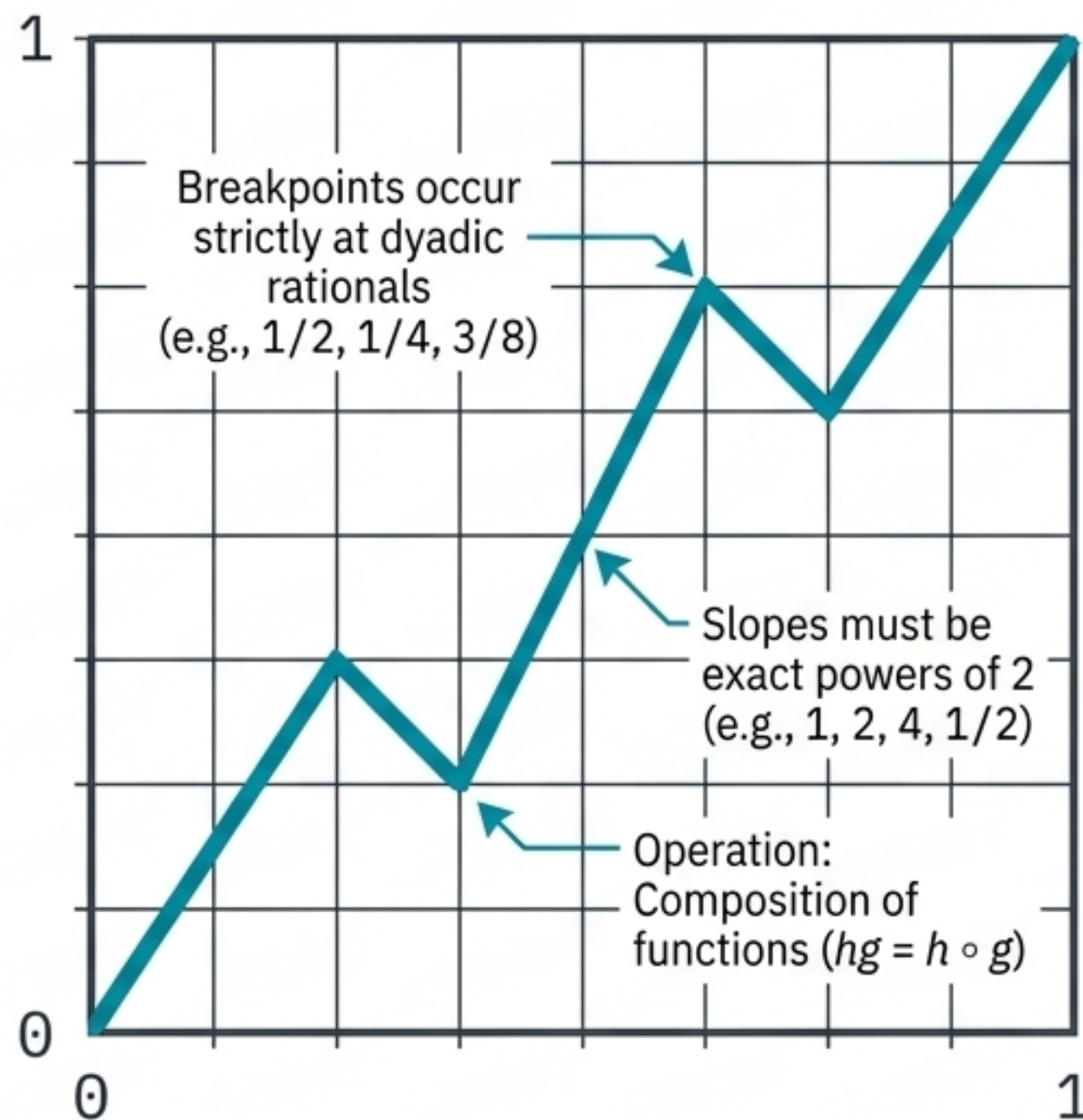
The von Neumann-Day Question: Must a **nonamenable** group contain a **free subgroup F_2** ?

1980

Ol'shanskii's **Tarski Monster** answers NO.

Enter Thompson's group F.

Introduced in 1965, group F consists of increasing, piecewise-linear bijections of $[0, 1]$. It is generated by a rigid set of dyadic bisections.



The ultimate test case defies classification.

Group F became infamous. It lacks the free subgroups that usually cause nonamenability, yet it falls outside the class of **elementary amenable** groups. Between 2009 and 2021, faulty proofs claiming both **amenable** and **nonamenable** were published and retracted.

	Finite Groups	Abelian Groups	Free Group (F_2)	Thompson's F
Left-Invariant Mean?	Yes	Yes	No	???
Contains Free Subgroup?	No	No	Yes	No (Brin-Squier, 1985)
Elementary Amenable?	Yes	Yes	No	No (Cannon-Floyd-Parry)

The resolution of Geoghegan's 1979 conjecture.

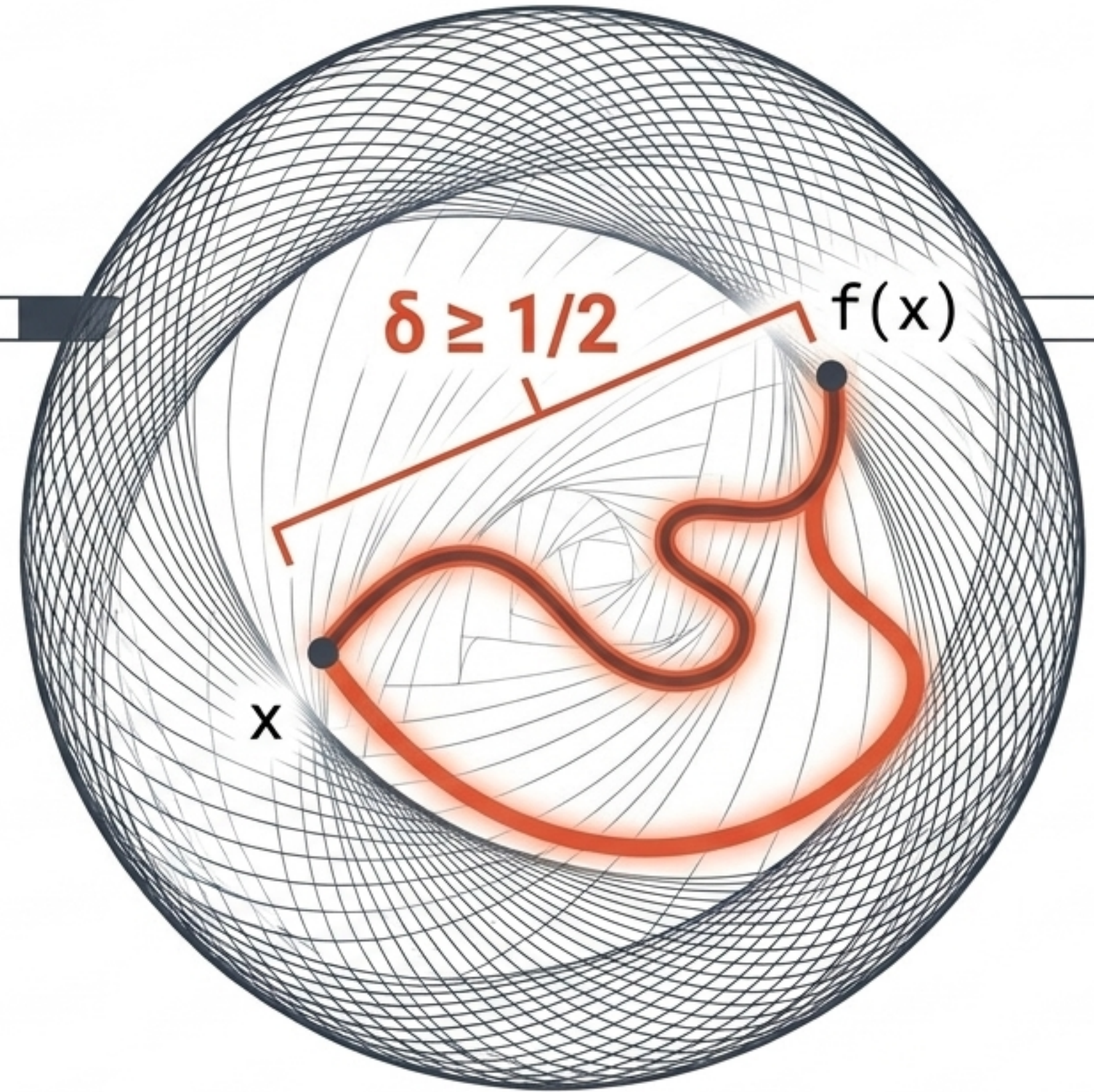
**Theorem 1.1: Thompson's
group F is **not amenable**.**

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Building a geometric trap using infinite dimensions.

The proof strategy relies on the **Benyamini-Sternfeld lemma**: there exists a **Lipschitz map f** on a Hilbert ball that moves **every single point** by at least a constant distance δ .

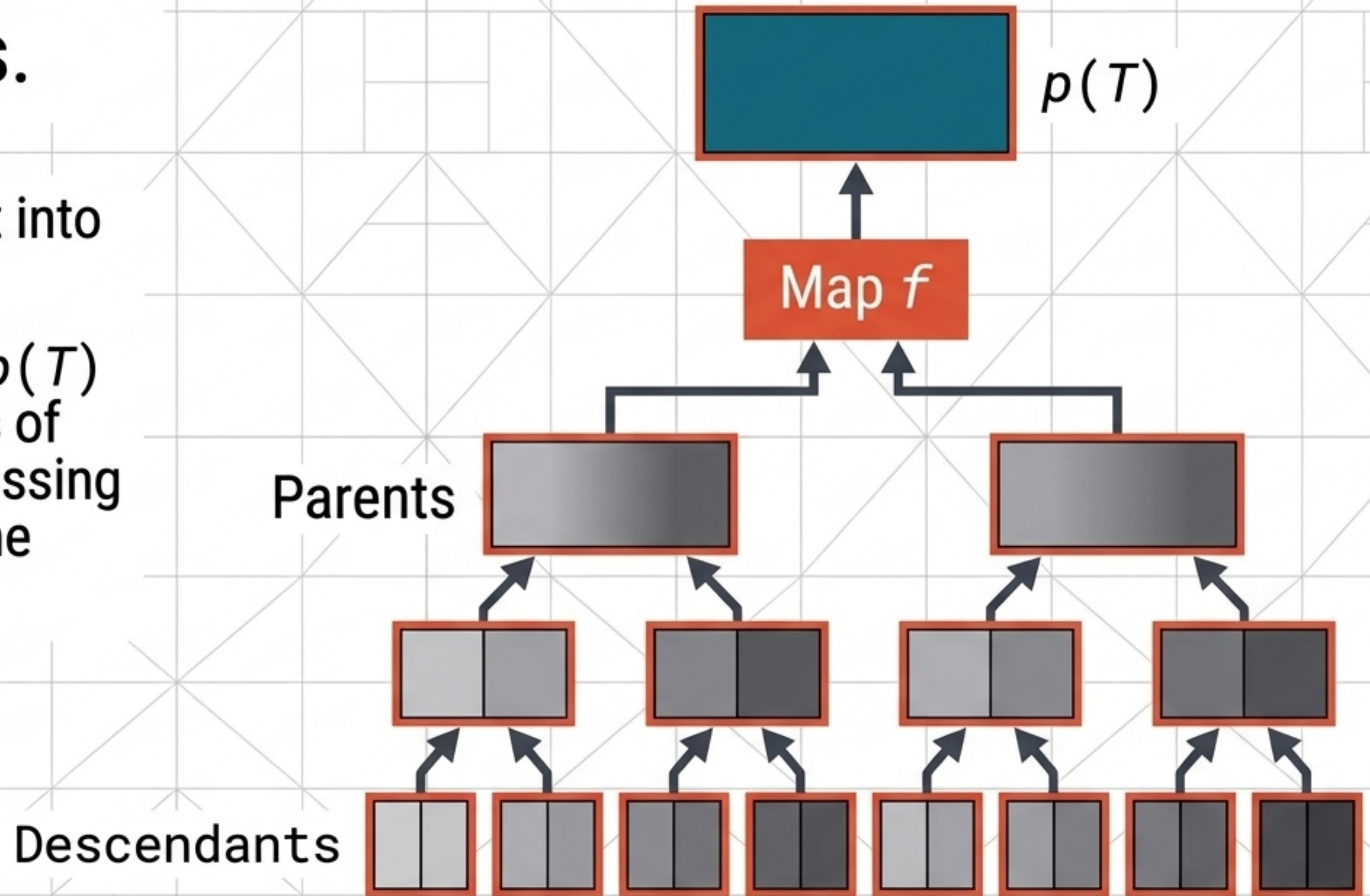
This unavoidable displacement creates **absolute mathematical friction**.



Recursive coloring of dyadic grids.

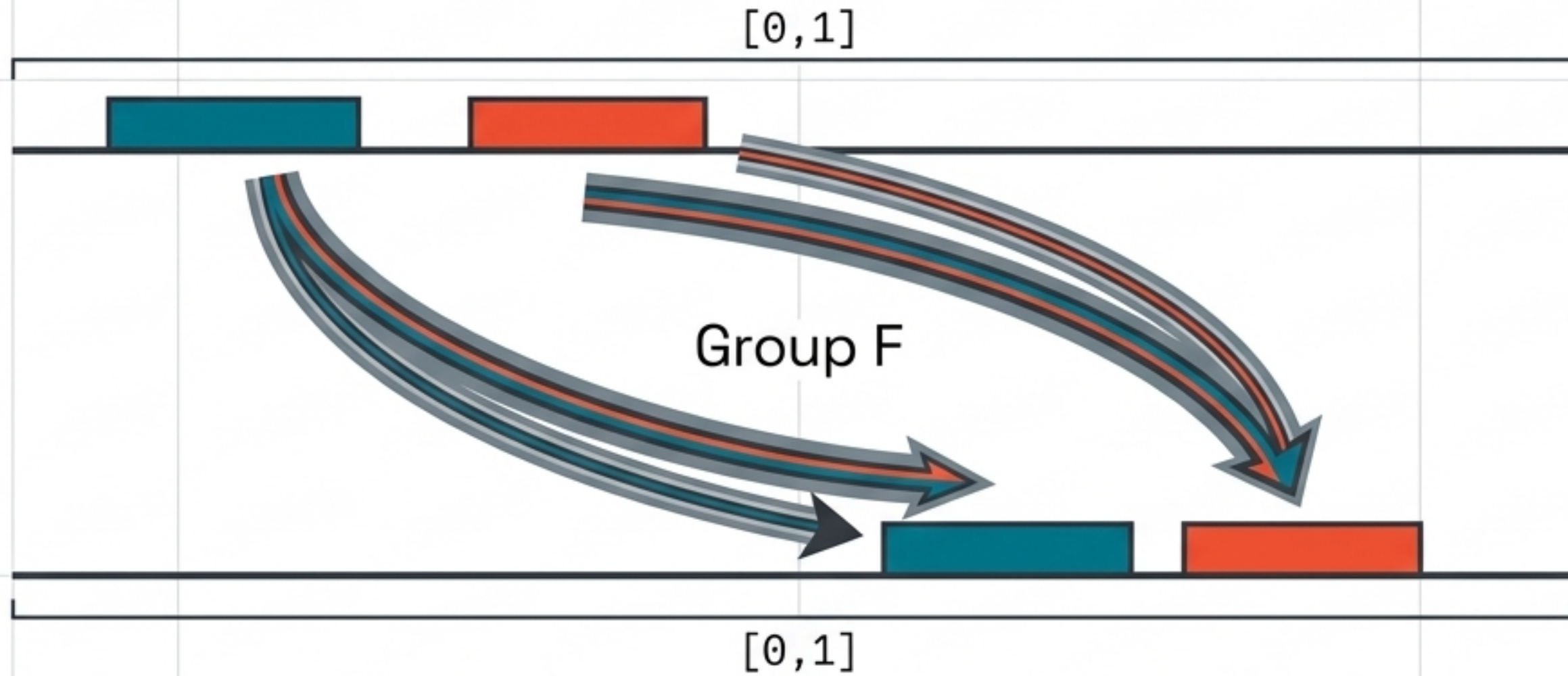
The unit interval is split into uniform dyadic grids.

Each grid gets a color $p(T)$ by averaging the colors of its descendants and passing that average through the displacement map f .



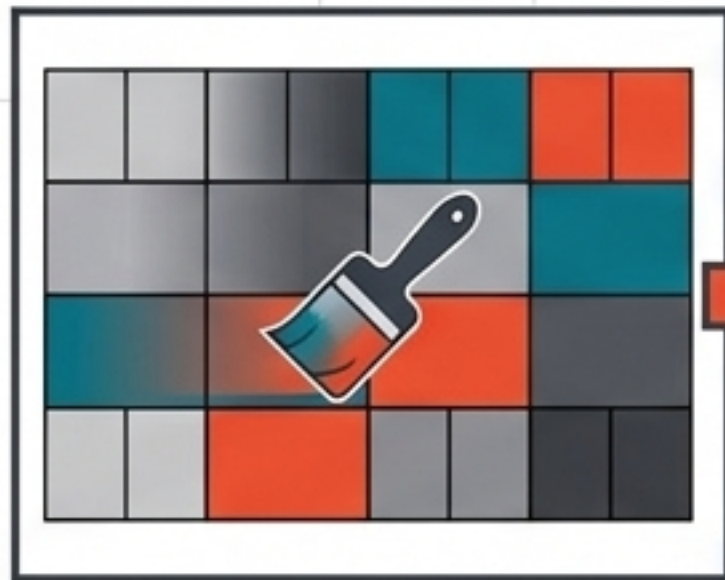
Exact affine transport and scalar correlation.

Group F can carry any separated pair of dyadic intervals onto any other. Therefore, averaging over a hypothetical Følner set forces all separated pairs to have nearly equal mean correlations, perfectly distributing the geometric friction.

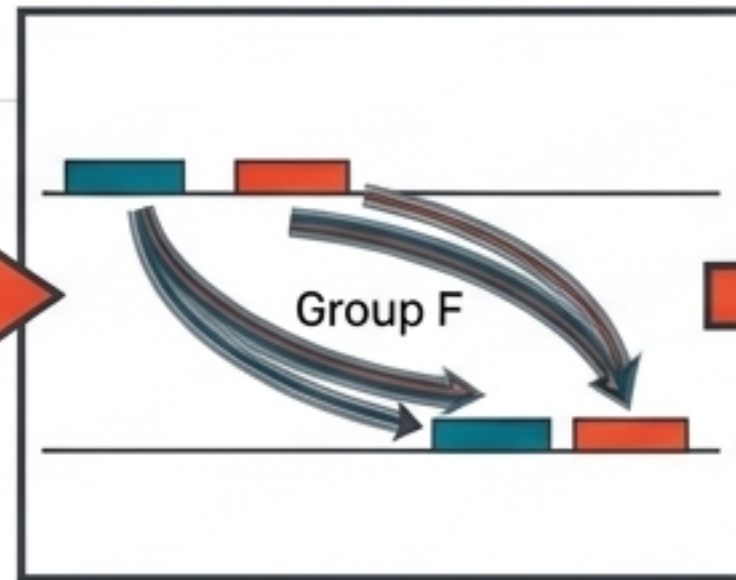


The mathematical contradiction.

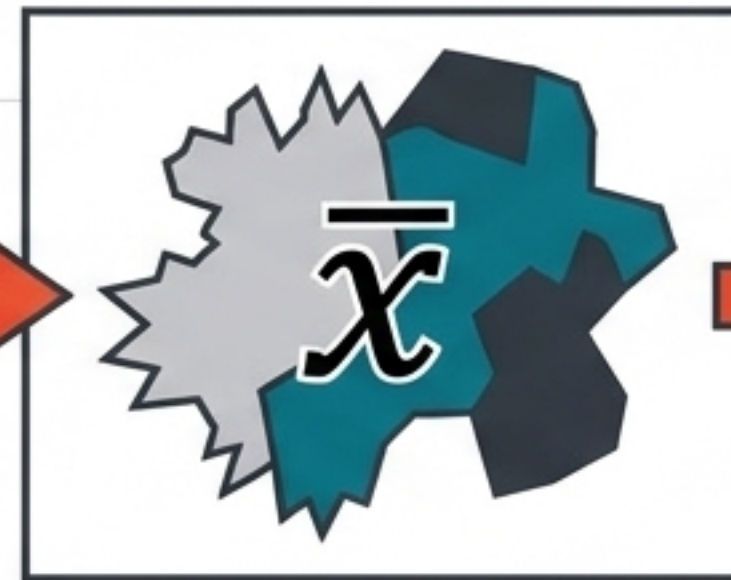
If a Følner set exists, the boundary η must be tiny. But the variance bound requires $\delta^2 \leq L^2(4/D + 4\eta)$. Because displacement δ is strictly bounded away from 0, η is forced to be large. The Følner condition shatters.



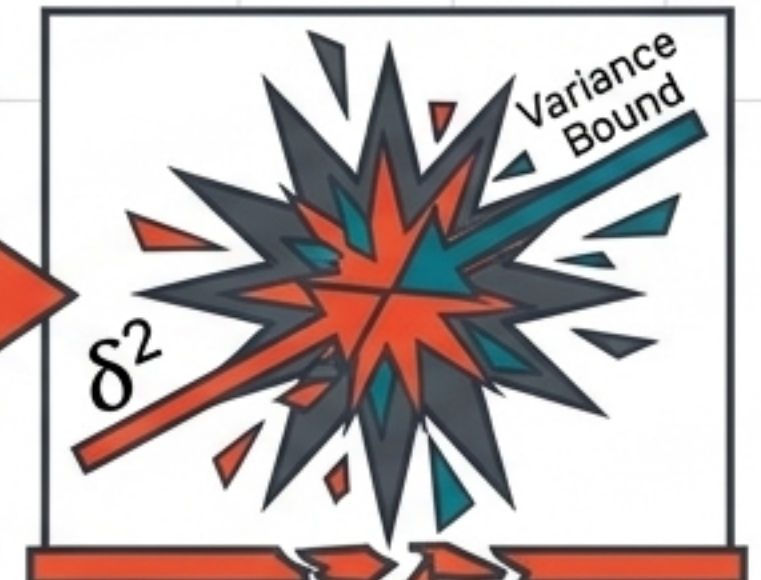
Map: Colorize intervals using displacement map f .



Transport: Use F to rearrange separated blocks.



Average: Average over a hypothetical Følner set.
Variance Bound:
 $\leq 4/D + 4\eta$

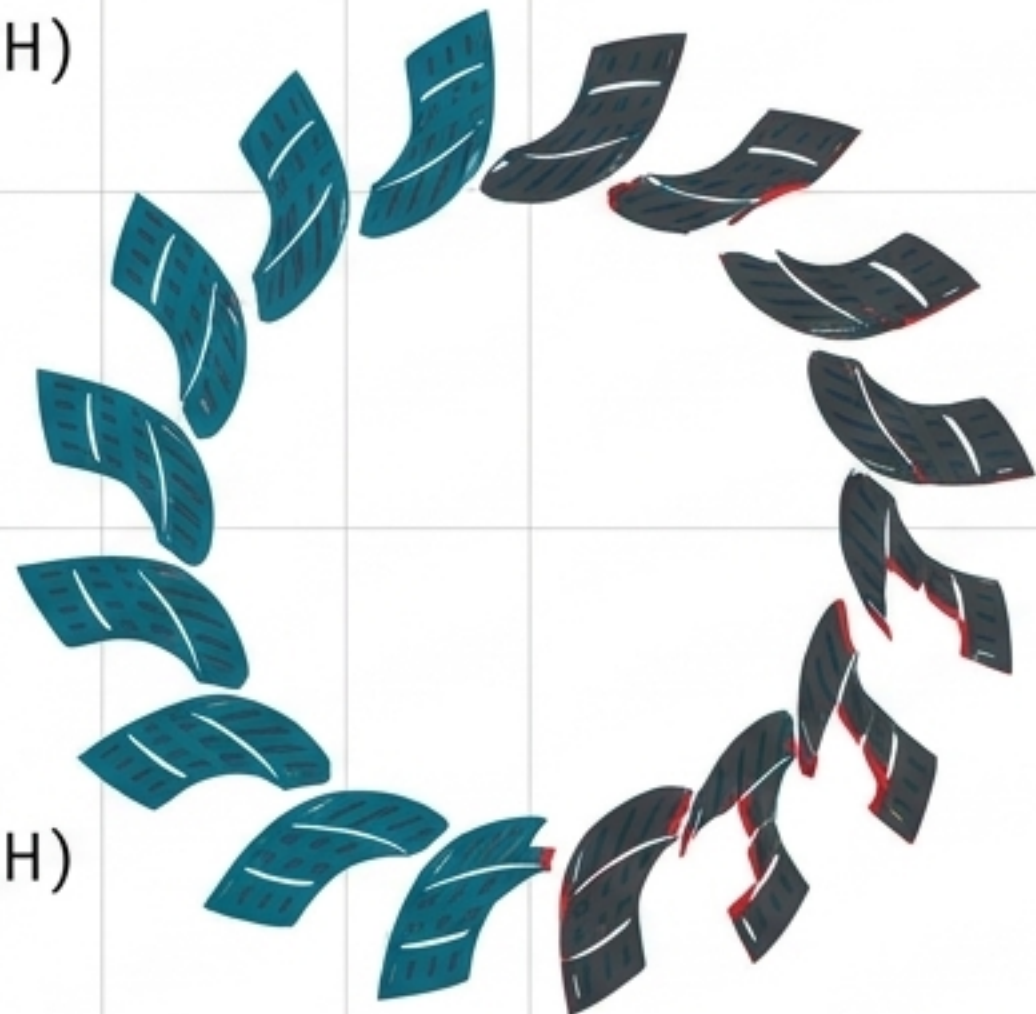


The Snap: Displacement bound (δ^2) collides with Variance Bound.

Domino effects in functional analysis and probability.

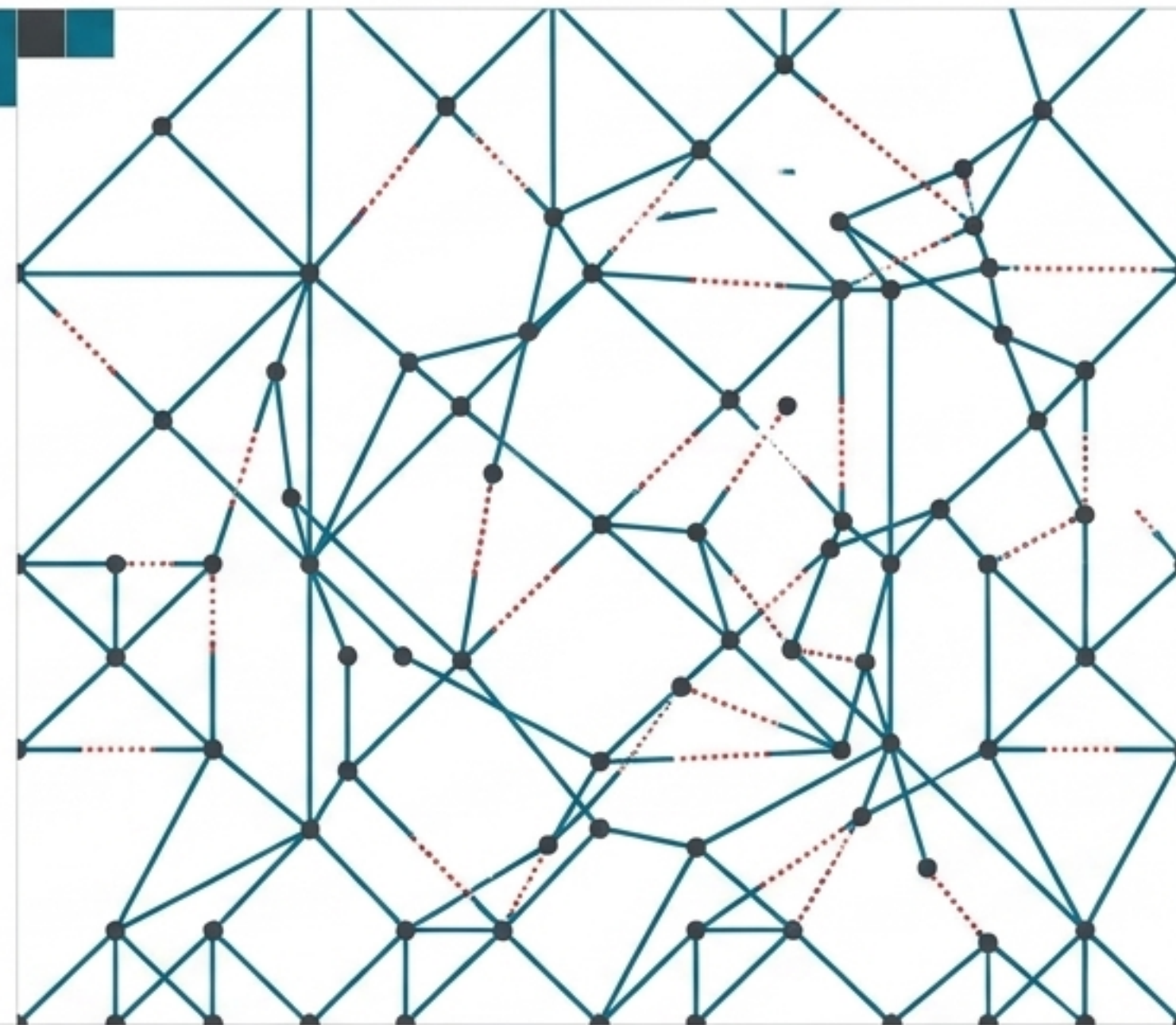
$GL(H)$

$GL(H)$



Representations

Proves there are uniformly bounded representations of $\mathbb{F}$ that cannot be made unitary by any bounded invertible operator.



Probability

Establishes nonuniqueness of percolation on the Cayley graphs of $\mathbb{F}$, guaranteeing infinitely many infinite open clusters between phase thresholds.

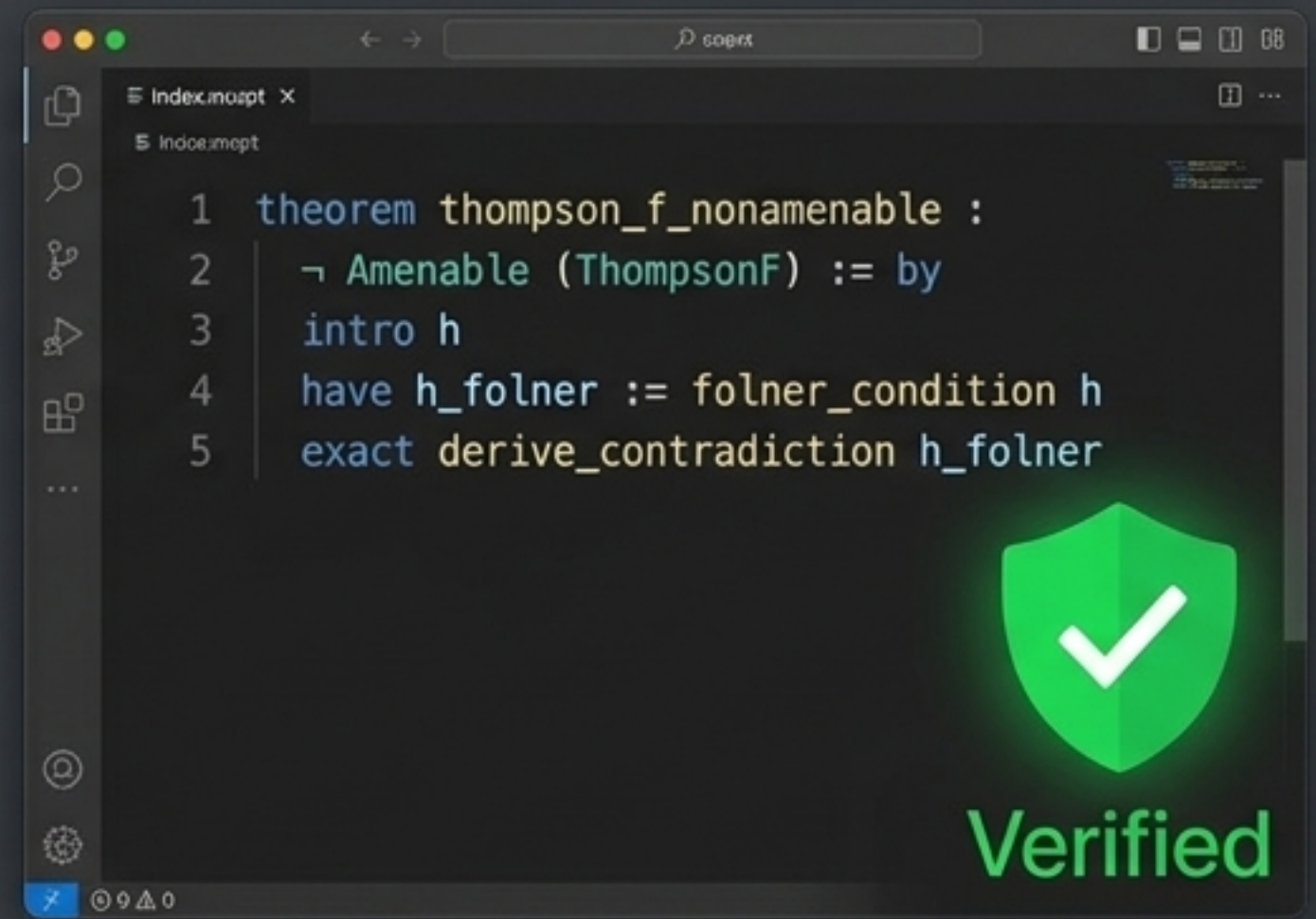
Human ingenuity meets machine verification.

A century of conjecture ends not just with a written paper, but with complete formal verification in the Lean 4 theorem prover.

Human

The absence of a positive normalized left-invariant mean for Thompson's group $\mathbb{F}$ is strictly proven.

Machine





The anomaly is resolved.

Thompson's group F stands as a fully verified, nonamenable entity—proving once and for all that the free group F_2 is not the sole architect of mathematical paradoxes.